API Documentation

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SAP Business Network Global Track and Trace | LBN

Public

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Introduction

This document describes the APIs provided by SAP Business Network Global Track and Trace that are useable by applications.

There are two basic types of API:

- **Public cross-tenant APIs**

  These are published by SAP in the SAP Business Accelerator Hub with documentation. They are mostly used by data contributors. In this guide, you can also find detailed instruction on how to use them.

- **Generated tenant-local interfaces**

  These are tenant-specific interfaces used by solution owners. They are also APIs, but differ based on the tenant.

→ Recommendation

Before you start working through this document, ensure that you have the most recent version of this document available from the SAP Help Portal at:

help.sap.com/gtt

Rate Limiting

This section describes the rate limit of SAP Business Network Global Track and Trace APIs.

Rate limiting restricts the number of incoming requests from callers to GTT APIs and within a specific time frame. It's a protective measure that:

- prevents the GTT service from crashing due to the excessive load of requests
- ensures the availability and optimal performance of the GTT service for every client.

We have defined the rate limits of following APIs according to tenant type:

API	Type of Tenant	Limit
IDOC interface: `GTT_V2_EM_IDOC_Ingestion` SOAP Message Interface: `TransportationOrderGenericTrackedProcessRequest_Out`	Production	The rate limit per tenant is 600 requests per minute.
	Test	The rate limit per tenant is 600 requests per minute, 10000 requests per day.
Write Service API: `GTT_V2_HTTP_EVENT_INGESTION`	Production	The rate limit per tenant is 600 per minute.
	Test	The rate limit per tenant is 150 requests per minute, 10000 requests per day.
Read Service API: `GTT_V2_OData_Query`	Production	The rate limit per tenant is 200 per minute.

API	Type of Tenant	Limit
Note This will take effective as of December 13, 2025. There is no limit on API calls to fulfillment tracking apps, such as the "Track Shipments" app.	Test	The rate limit per tenant is 60 requests per minute.

When the rate limit is exceeded, you will receive an `HTTP 429 Too Many Requests` response status code, and your requests are rejected. You can try executing requests again several minutes later.

API Payload Limits

To prevent APIs from being overloaded by the payload size, we have introduced API payload limits to regulate the API usage per message.

The following tables list the size limit of different components in API payload:

- Public Tracking APIs

API	Request Body	Response Body
Provider API for Order Events	Less than 40MB	N/A
Consumer API for Order Tracking	Less than 40MB	N/A

- Tenant-local Interfaces

API	Request Body without Attachment	Attachment	Response Body
IDOC Service	Less than 15MB	Not supported	N/A
Read Service API	No limit	N/A	The response body of read service API supports: ◦ 256 records at maximum for main entity in a query; ◦ 5000 records at maximum for each expanded entity of a single main entity; ◦ 20000 records at maximum for sum of each expanded entity for all main entity.
Write Service API	Less than 2MB	Less than 10MB	N/A
SOAP Message Interface	Less than 15MB	Not supported	N/A

Here are examples of requests and responses for you to better understand the Read Service API size limit:

Example

Request	Result	Odata Response
`odata/v1/com.sap.gtt.app.gttft1.gttft1Service/Shipment?filter=filter1&$expand=processEvents, plannedEvents, freightUnitTPs`	1 shipment was found and 5100 process events found on this shipment	**`Error: The response body`**
`odata/v1/com.sap.gtt.app.gttft1.gttft1Service/Shipment?filter=filter1&$expand=processEvents, plannedEvents, freightUnitTPs`	10 shipments were found and 2100 process events were found on each shipment.	**`Error: The response body`**
`odata/v1/com.sap.gtt.app.gttft1.gttft1Service/Shipment?filter=filter1&$expand=processEvents, plannedEvents, freightUnitTPs,stops,trackingUnitTPs`	1 shipment was found with 5000 process events, 5000 planned events, 5000 freightUnitTPs, 5000 stops, and 5000 trackingUnitTPs.	**`Successful with data`**

Public Cross-Tenant Tracking APIs

This chapter describes how solution owners and data contributors exchange messages via the tracking APIs (or tracking interfaces) offered by SAP Business Network Global Track and Trace.

SAP Business Network Global Track and Trace provides two types of tracking APIs:

- **Consumer API for Order Tracking (GTT)**: enables solution owners to send a tracking request for an order to data contributors.
- **Provider API for Order Events (GTT)**: enables data contributors (such as network partners or carriers) to post events for a tracked order.

You can check out the above APIs on SAP Business Accelerator Hub.

This chapter specifically describes the following two cases:

- how a solution owner sends a tracking request to a data contributor using the **Consumer API for Order Tracking (GTT)**,
- and how a data contributor sends back the tracking data (that will later be converted to and processed as events in the GTT system) to a solution owner using the **Provider API for Order Events (GTT)**.

Sending Tracking Requests via Consumer API for Order Tracking (GTT)

As a solution owner, you can send a tracking request to a data contributor using the Consumer API for Order Tracking (GTT).
This chapter contains the following sections:

- Prerequisites
- Configuration
- Testing the Integration

Prerequisites

To perform the tasks described in this chapter, you, as a solution owner, must have the following prerequisites:

1. You have onboarded on SAP Business Network for Logistics.
2. You have sent the invitation to the data contributor using the Invite Business Partners app. After the data contributor accepts the invitation using the Manage Invitations app, you two will be connected as business partners.
3. You have established a B2B connection to the external systems of the data contributor in the Configure Partner Connections app.
4. You have ensured that the data contributor has completed the Prerequisites and Configuration.

Related Information

Invite Business Partners
Configure Partner Connections

Configuration

Context

In the Manage Models app of SAP Business Network Global Track and Trace, the visibility provider integration information is already maintained for you in the GTT standard model “gttft1”.

But if you want to integrate with your visibility provider in a user model or map user-defined fields in the standard model, do the following:

Procedure

1. On the SAP Fiori launchpad, open the Manage Models app.
2. Select a model.
3. Go to the **Visibility Provider Integration** tab and open the edit mode.
4. Select a tracked process, and then turn on the **Integration Switch**.
5. In the **Tracked Process / Events** list on the left, you can see the tracked process event and event types. Use the value help to fill in the SAP Business Network for Logistics event type for actual events.
6. Select an event in the list, and then use the value help to fill in the fields or SAP Business Network for Logistics tracking fields on the right.
7. Save and deploy the model.

Testing the Integration

This section contains the following subsections:

- Procedures
- Sample Payload

Procedures

Prerequisites

You must complete the procedures below to note down the data contributor's identifier that will later be used in the sample payload to fill in the `serviceAgentLbnId` field.

On the SAP Fiori launchpad, open the ***Configure Partner Connections*** app, choose a partner connection that is active.

- If the data contributor you chose is a carrier, then the carrier must be onboarded onto SAP Business Network for Logistics. Note down its **LBN ID** with an “LBN#” prefix, such as "LBN#1234567".
- If the data contributor you chose is a network partner, then its corresponding carrier doesn’t have to be onboarded onto SAP Business Network for Logistics. Note down its **Business Partner ID Type** and **Business Partner ID Value**, with a hash “#” in between, such as "SCAC#MAEU".

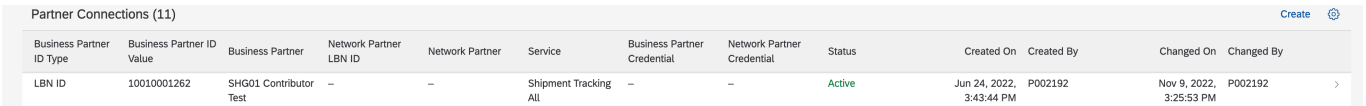
ℹ Note

You can use the table below for reference.

Business Partner ID Type	Correct Form
LBN ID	`LBN#`
SCAC	`SCAC#`
P44_EU	`P44_EU#`
Motor Carrier Number	`MC#`
US DOT	`USDOT#`
GLN	`GLN#`
VAT	`VAT#`
SAP Business Partner Number	`SAPBPID#`

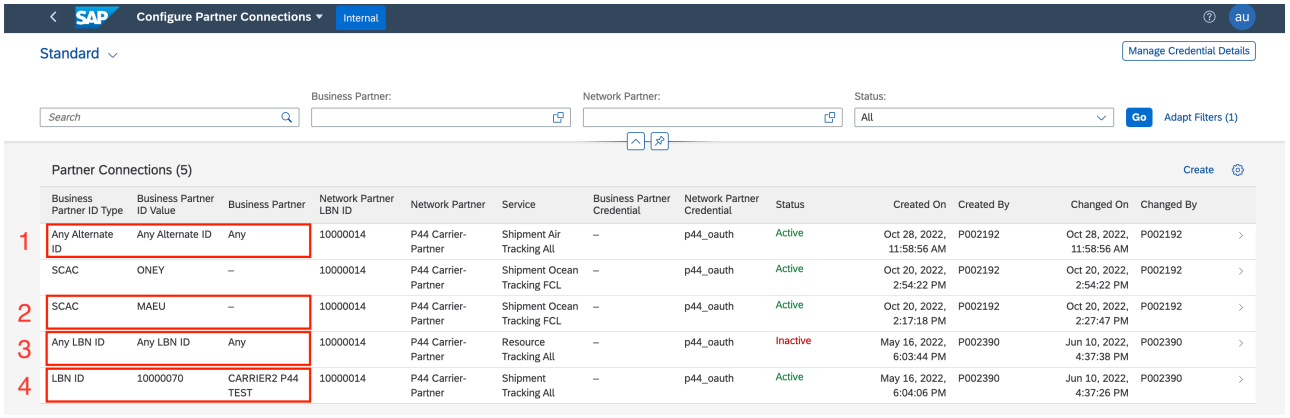
You can refer to the screenshots below for different scenarios:

- The data contributor is a carrier:



- The data contributor is a network partner:
  - Connection 1 is for any partner who is not onboarded.
  - Connection 2 is for a partner with a specific carrier identifier who is not onboarded.
  - Connection 3 is for any partner who is onboarded.

- Connection 4 is for a partner with a specific carrier identifier who is onboarded.



Steps

To test the integration in SAP Business Network Global Track and Trace, carry out the steps below:

1. On the SAP Fiori launchpad, open the Manage Models app.
    a. Select the GTT standard model “gttft1”.
    b. Switch to the **Runtime** view on the top right.
    c. In the **Write Service** tab, find the **ShipmentEvent**. Click the green section of this API endpoint.
    d. Click **Try it out** to test the API.
    e. In the **Request Body** text box, clear the existing request payload. Copy and paste the Sample Payload we provided and customize it to fit your needs.
    f. Execute the request.
    g. In the **Responses** section, it shows the response with the endpoint.
2. (Option 1) Back to the SAP Fiori launchpad, open the View Logs app. Open the Manage Message Logs app from the **Quick Links** section to check out the message processing status:
    a. Select the latest message on the list with the source type as **External**. The message contains the tracking request you just sent. You can drill down for more information.
    b. Back to the worklist page, refresh the page until you see a new message from your visibility provider. If the status is **Success**, then you have sent the tracking request successfully.
3. (Option 2) Back to the SAP Fiori launchpad, open the View Logs app. Open the Manage Integration Logs – Version 2.0 app from the **Quick Links** section to check out the message processing status. For instructions on how to use the app, open the in-app help by selecting ⓘ within the **Manage Integration Logs – Version 2.0** app.

Sample Payload

The sample payload is provided for you to test the integration. You can refer to the following table for instructions on how to provide values for some key parameters in the payload:

Parameter	Value
`altKey`	Mandatory. Fill the alternative key in the following pattern: *<scheme>*:*<party>*:*<system>*:*<type>*:*<id>*. For example,

Parameter	Value
	"xri://sap.com/id:LBN#10000000001:QX8CLNT912:FT1_SHIPMENT:6100079522". For more, see Process List.
`actualBusinessTimestamp`	Mandatory. Fill in the current timestamp in UTC time format, for example, "2022-07-01T00:00:00Z".
`shippingType_code`	Mandatory. Refer to the "Shipping Type Descriptions" section of the Code Lists document.
`serviceAgentLbnId`	Fill in the data contributor's identifier that you noted down in the prerequisite of Procedures.
`transportationMode_code`	Mandatory. Refer to the "Transportation Modes" section of the Code Lists document.
`plannedEvents`	Provide at least one planned Departure event and one planned Arrival event, with their location alternative keys maintained in the Manage Locations app.
`stops`	Make sure all the locations mentioned in planned events are described here.
`idType_code` of `trackedObjects`	Only one tracked object is supported. Refer to the "Tracked Object Type" section of the Code Lists document.
`docType_code` of `carrierRefDocuments`	Refer to the "Shipper/Carrier Reference Document Type Codes" section of the Code Lists document.

The basic code structure of the sample payload is provided below. Please customize it to fit your needs.

Sample Code


```
{
  "altKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:FT1_SHIPMENT:6100079522",
  "actualTechnicalTimestamp": "2024-12-23T09:16:03+01:00",
  "actualBusinessTimestamp": "2024-12-23T09:16:03+01:00",
  "actualBusinessTimeZone": "CET",
  "references": [],
  "plannedEvents": [
    {
      "eventType": "com.lbngttsampletest.gtt.app.gttft1.Shipment.LoadingStart",
      "locationAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:Location:LogisticLocation:V
      "plannedTechnicalTimestamp": null,
      "plannedTechTsEarliest": null,
      "plannedTechTsLatest": null,
      "plannedBusinessTimestamp": "2024-12-23T00:00:00+01:00",
      "plannedBizTsEarliest": null,
      "plannedBizTsLatest": null,
      "plannedBusinessTimeZone": "CET",
      "eventMatchKey": "61000795220001",
      "payloadSequence": 1
    },
    {
      "eventType": "com.lbngttsampletest.gtt.app.gttft1.Shipment.LoadingEnd",
      "locationAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:Location:LogisticLocation:V
      "plannedTechnicalTimestamp": null,
      "plannedTechTsEarliest": null,
      "plannedTechTsLatest": null,
      "plannedBusinessTimestamp": "2024-12-23T00:00:00+01:00",
      "plannedBizTsEarliest": null,
      "plannedBizTsLatest": null,
```

```
      "plannedBusinessTimeZone": "CET",
      "eventMatchKey": "61000795220001",
      "payloadSequence": 2
    },
    {
      "eventType": "com.lbngttsampletest.gtt.app.gttft1.Shipment.POPU",
      "locationAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:Location:LogisticLocation:V
      "plannedTechnicalTimestamp": null,
      "plannedTechTsEarliest": null,
      "plannedTechTsLatest": null,
      "plannedBusinessTimestamp": "2024-12-23T00:00:00+01:00",
      "plannedBizTsEarliest": null,
      "plannedBizTsLatest": null,
      "plannedBusinessTimeZone": "CET",
      "eventMatchKey": "61000795220001",
      "payloadSequence": 3
    },
    {
      "eventType": "com.lbngttsampletest.gtt.app.gttft1.Shipment.Departure",
      "locationAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:Location:LogisticLocation:V
      "plannedTechnicalTimestamp": null,
      "plannedTechTsEarliest": null,
      "plannedTechTsLatest": null,
      "plannedBusinessTimestamp": "2024-12-23T00:00:00+01:00",
      "plannedBizTsEarliest": null,
      "plannedBizTsLatest": null,
      "plannedBusinessTimeZone": "CET",
      "eventMatchKey": "61000795220001",
      "payloadSequence": 4
    },
    {
      "eventType": "com.lbngttsampletest.gtt.app.gttft1.Shipment.Arrival",
      "locationAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:Location:LogisticLocation:S
      "plannedTechnicalTimestamp": null,
      "plannedTechTsEarliest": null,
      "plannedTechTsLatest": null,
      "plannedBusinessTimestamp": "2024-12-23T00:00:00+01:00",
      "plannedBizTsEarliest": null,
      "plannedBizTsLatest": null,
      "plannedBusinessTimeZone": "CET",
      "eventMatchKey": "61000795220002",
      "payloadSequence": 5
    },
    {
      "eventType": "com.lbngttsampletest.gtt.app.gttft1.Shipment.UnloadingStart",
      "locationAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:Location:LogisticLocation:S
      "plannedTechnicalTimestamp": null,
      "plannedTechTsEarliest": null,
      "plannedTechTsLatest": null,
      "plannedBusinessTimestamp": "2024-12-23T00:00:00+01:00",
      "plannedBizTsEarliest": null,
      "plannedBizTsLatest": null,
      "plannedBusinessTimeZone": "CET",
      "eventMatchKey": "61000795220002",
```

```
        "payloadSequence": 6
      },
      {
        "eventType": "com.lbngttsampletest.gtt.app.gttft1.Shipment.UnloadingEnd",
        "locationAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:Location:LogisticLocation:S
        "plannedTechnicalTimestamp": null,
        "plannedTechTsEarliest": null,
        "plannedTechTsLatest": null,
        "plannedBusinessTimestamp": "2024-12-23T00:00:00+01:00",
        "plannedBizTsEarliest": null,
        "plannedBizTsLatest": null,
        "plannedBusinessTimeZone": "CET",
        "eventMatchKey": "61000795220002",
        "payloadSequence": 7
      },
      {
        "eventType": "com.lbngttsampletest.gtt.app.gttft1.Shipment.POD",
        "locationAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:Location:LogisticLocation:S
        "plannedTechnicalTimestamp": null,
        "plannedTechTsEarliest": null,
        "plannedTechTsLatest": null,
        "plannedBusinessTimestamp": "2024-12-23T00:00:00+01:00",
        "plannedBizTsEarliest": null,
        "plannedBizTsLatest": null,
        "plannedBusinessTimeZone": "CET",
        "eventMatchKey": "61000795220002",
        "payloadSequence": 8
      }
    ],
    "shipmentNo": "6100079522",
    "serviceAgentLbnId": "LBN#10000000002",
    "transportationMode_code": "03",
    "dangerousGoods": false,
    "shippingType_code": null,
    "carrierRefDocuments": [
      {
        "docId": "AVB12345678",
        "docType_code": "T50"
      },
      {
        "docId": "KV9876",
        "docType_code": "BN"
      }
    ],
    "trackedObjects": [
      {
        "idType_code": "CONTAINER_ID",
        "value": "HB12356"
      }
    ],
    "trafficDirection_code": null,
    "incoterms_code": "CFR",
    "incotermLocation": "GERMANY",
    "plannedTotalDistance": 278.298727,
```